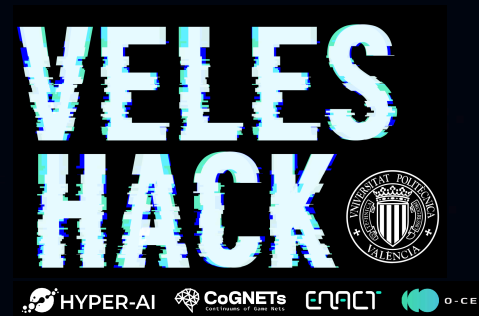


Sunflower

Moves a running app to the computer with the cleanest power, without losing a request.

Challenge 3, ENACT: Kubernetes Dynamic Adaptation



Funded by
the European Union



GitHub repo

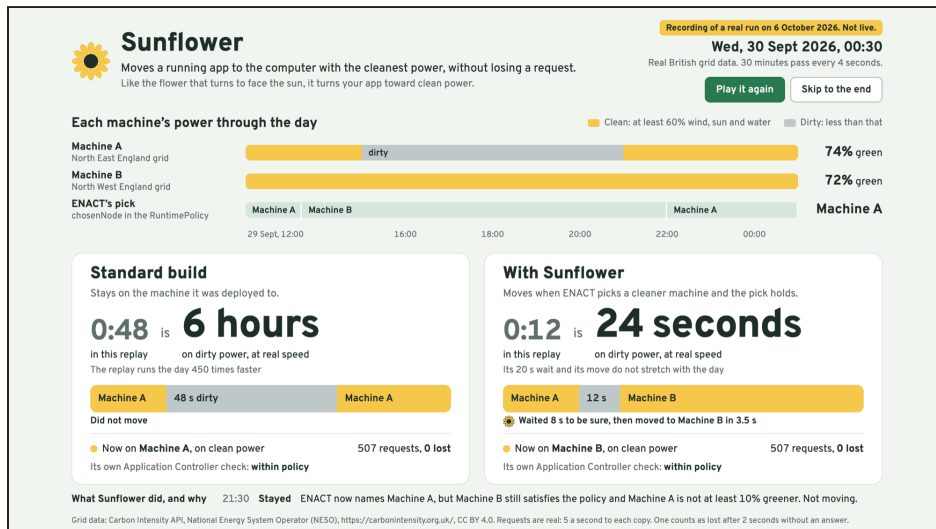
GitHub repo: github.com/kinnuworks/sunflower-enact

- **Start with the README:** the result, the challenge checklist with a screenshot of every SDK step, and how to run it.
- **A recorded run you can play in a browser:** docs/demo, drawn from the measured data, unedited.
- **The raw results:** evidence/ holds the file behind every number I quote.
- **What I am giving back to ENACT:** docs/field-report.md, 45 findings, each with the fix I used.
- Licence: Apache-2.0



Summary

- ENACT's policy operator names the best machine for an app and keeps that answer current as the grid changes.
- **After deploy day the app does not follow.** The SDK pins it once, so when its machine's power turns dirty, GreenCharge stays there, below its own 60% rule.
- **Sunflower is a small controller that reads ENACT's choice and moves the app to it,** once the change has lasted, without losing a request.
- I completed all five items of the challenge checklist in the ENACT SDK, dataspace included, then built Sunflower on top.



One real run on the challenge cluster. Left: the standard build, 6 hours on dirty power at real speed. Right: with Sunflower, about 24 seconds. Each was sent 507 requests and lost none.



Highlights

0

requests lost

across 30 moves in a row, 1,777 requests, on the challenge cluster

6 h → 24 s

on dirty power, at real speed

standard build against Sunflower, over a real day of British grid data

45

findings given back

problems I hit in ENACT, each written up with the fix I used

- **How I get 6 hours against 24 seconds:** my replay runs the day 450 times faster and shows 48 s against 12 s. The standard build's 48 s are the whole dirty stretch, which is 6 hours at real speed. Sunflower's 12 s are a wait and a 3.5 s move that do not stretch with the day: with its default 20 s wait, about 24 seconds.
- **Checklist done in the ENACT SDK:** dataspace feed, policy model, Application Controller extension (14 tests), policy, packaging, deployment and monitoring, plus one step with the SDK's assistant. The repo has a screenshot of every step.
- **ENACT stays in charge:** Sunflower reads the policy's chosen machine and has no ranking of its own. It waits out short dips, ignores near-ties, limits how often it moves, and undoes a move that fails.
- **What I do not claim:** carbon saved. This cluster reports energy as a model estimate, so I report time on dirty power, which I measured.